

# HW 5 theory questions

## HW 5 Q5

a)  $.2 + .3 - .1 = .4$

b)  $1 - .4 = .6$

c) NO, because  $P(A) \cdot P(B) \neq P(A \text{ and } B)$

d) NO, because  $P(A \cap B) = .01$  and mutually exclusive events can't happen at the same time

## HW 5 Q6

a)  $33/100$  or  $.33$

b)  $0.5$  or  $\frac{50}{100}$

c)  $(.5) \cdot (.33) = .17$

d)  $\left(\frac{.17}{.5}\right) = .34$

## HW 5 Q8

a)  $(.25) \cdot (.6) = 0.15$

b)  $(.4) \cdot (.8) = 0.32$

c)  $(0.15) + (0.32) + (0.09) = 0.56$

## HW 5 Q7

a)  $15/180 = 0.83$

b)  $55/180 = 0.361$

c)  $30/180 = 0.166$

d)  $(15/180) \cdot (55/180) = 0.27$